Context-Aware Explainable Deep Learning for Suicidal Ideation Detection from Social Media Captions

1st Rohit Hazra
Department of Computer Science & Engineering
Balasore College of Engineering & Technology
Balasore, India
l.rohit.hazra@gmail.com

2nd Tanmay Mohanty
Department of Computer Science & Engineering
Balasore College of Engineering & Technology
Balasore, India
mohantytanmay396@gmail.com

3rd Rashmita Mohalik
Department of Computer Science & Engineering
Balasore College of Engineering & Technology
Balasore, India
mohaliknitu123@gmail.com

4th Diptiranjan Behera
Department of Computer Science & Engineering
Balasore College of Engineering & Technology
Balasore, India
beheradiptiranjan435@gmail.com

5th Shubhranshu Sahoo
Department of Computer Science & Engineering
Balasore College of Engineering & Technology
Balasore, India
shubhranshusahoo98@gmaail.com

***Abstract* - Digital mental health surveillance offers a vital window into acute psychological distress, yet informal syntax, colloquial expressions, and high inference latency hinder early suicide risk detection on social media. This paper presents a context-aware, explainable deep learning framework that couples a lightweight transformer with recurrent sequential modeling and attention pooling. The architecture utilizes a pre-trained DistilRoBERTa encoder to generate dense 768-dimensional token representations from noisy text, followed by a Bidirectional Gated Recurrent Unit (BiGRU) to capture the chronological unfolding of emotional intent across user disclosures. A scaled dot-product context-attention mechanism dynamically filters conversational fillers, weighting subtle crisis cues to synthesize an emotionally dense sequence representation. Evaluated on a balanced 20,000-post corpus derived from the Kaggle Suicide and Depression Detection benchmark, the model achieves 98.80% accuracy, precision, recall, and macro F1-score, paired with an ROC-AUC of 0.9986 on a held-out test partition of 3,000 posts. This performance notably surpasses standard DistilRoBERTa, BERT-base, and BiLSTM baselines. Integrating attention heatmaps and Integrated Gradients delivers token-level interpretability, ensuring clinical accountability alongside high computational efficiency for real-time digital triage.**

Keywords: Suicidal Ideation Detection, DistilRoBERTa, BiGRU, Context Attention, Explainable AI (XAI), Integrated Gradients, Social Media Surveillance.

I. INTRODUCTION

Behind the constant flow of social media feeds, online platforms have become an unintended refuge where individuals openly express severe psychological distress, hopelessness, and suicidal thoughts long before seeking professional care. Suicide remains a critical global public health concern, claiming roughly one million lives each year and disproportionately affecting youth and young adults. Because early detection can prevent tragic outcomes, automated surveillance of online disclosures provides an indispensable opportunity for proactive intervention. However, deciphering digital writing is challenging; social media posts are inherently unstructured, ranging from brief, slang-heavy captions to sprawling personal narratives laced with informal abbreviations, subtle metaphors, and emotional nuances that standard text filters struggle to capture. [1]

Traditional detection methods rely on rigid keyword searches and frequency-based models that flag isolated terms like "kill," "die," or "lost". These non-contextual approaches frequently trigger false alarms on casual venting while missing quiet, implicit cries for help that lack explicit distress vocabulary. On the other hand, while large transformer models like BERT provide rich contextual awareness, their heavy parameter footprints cause high inference latency and memory strain, making continuous real-time surveillance difficult to deploy. Moreover, standard deep learning models act as opaque black boxes that output a single risk score without explaining why a user was flagged, creating a trust barrier for human reviewers, helpline operators, and clinical responders who require transparent evidence before initiating triage.[2]

To address these hurdles of contextual nuance, operational speed, and clinical accountability, this paper introduces an end-to-end, context-aware, and explainable deep learning framework for suicidal ideation detection. The architecture pairs a lightweight pre-trained DistilRoBERTa encoder—cutting model parameters by 40% for rapid inference while preserving robust subword comprehension—with a Bidirectional Gated Recurrent Unit (BiGRU) to track how psychological despair unfolds across a narrative. A scaled dot-product context-attention layer dynamically elevates

high-risk indicators while filtering irrelevant conversational filler. To overcome the black-box dilemma, the pipeline integrates attention heatmaps and Integrated Gradients for token-level decision rationales. Evaluated on a balanced dataset of 20,000 social media posts, the proposed framework achieves 98.80% across accuracy, precision, recall, and macro F1-score with an ROC-AUC of 0.9986, establishing an efficient, dependable, and interpretable foundation for digital mental health support.

II. LITERATURE REVIEW

Paper	**Proposed Model**	**Metrics (Accuracy, Precision, Recall, F1-Score)**	**Datasets Used**	**Remarks & Limitations Addressed by Proposed Model**
Tadesse et al. (2019) (Algorithms, MDPI) [3]	Stacked LSTM-CNN with Word2Vec word embeddings	Acc: 93.92% Prec: 93.90% Rec: 93.80% F1: 93.85%	Kaggle / Reddit r/SuicideWatch Benchmark Corpus	Static Word2Vec misses slang, misspellings and polysemy; no attention or explainability.
Ji et al. (2021) (IEEE TCSS)[4]	Pre-trained BERT-base with BiLSTM classifier	Acc: 93.70% Prec: 93.10% Rec: 93.50% F1: 93.30%	Kaggle Suicide and Depression Detection (Suicide_Detection.csv)	BERT has high latency; standard pooling may dilute crisis cues; lacks explainability.
Aldhyani et al. (2022) (MDPI IJERPH)[5]	Hybrid CNN-BiLSTM with Word2Vec / TF-IDF features	Acc: 95.00% Prec: 94.80% Rec: 95.10% F1: 94.95%	Kaggle Suicide and Depression Detection (Suicide_Detection.csv)	Word-level OOV issues; no attention for long disclosures; limited transparency.
Azhar et al. (2026)*(MDPI Information)*[6]	CrisisFormer (DistilBERT + Dense Head + SHAP/LIME)	Acc: 96.25% Prec: 96.30% Rec: 96.25% F1: 96.25%	Reddit *r/SuicideWatch* Corpus (40k posts)	[CLS] pooling may dilute sparse cues; limited sequential modeling.
Bajaj et al. (2025) *(MECS Press IJISA)* [7]	Fine-Tuned BERT-base vs. LSTM / XGBoost	Acc: 97.25% Prec: 97.26% Rec: 97.25% F1: 97.25%	Kaggle Suicide and Depression Detection (Suicide_Detection.csv)	BERT-base has higher computational cost; limited sequential modeling.
Proposed Model	DistilRoBERTa + BiGRU + Context Attention (with Integrated Gradients)	Acc: 98.80% Prec: 98.80% Rec: 98.80% F1: 98.80%	Kaggle Suicide and Depression Detection (Suicide_Detection.csv; 20,000 subset)	Combines subword robustness, sequential context, attention-based cue selection, and Integrated Gradients for explainability.

III. RESEARCH GAP:

Despite significant progress in suicidal ideation detection, existing approaches still show limitations in jointly addressing contextual understanding, sequential dependency modeling, attention-based cue selection, and explainability. Word-based methods such as Word2Vec and TF-IDF are affected by out-of-vocabulary words, slang, misspellings, and limited contextual representation. Hybrid CNN-LSTM/BiLSTM models improve sequential feature extraction but generally lack adaptive attention mechanisms for identifying the most informative crisis-related cues. Transformer-based approaches such as BERT provide stronger contextual representations but may introduce higher computational requirements, while standard classification architectures do not explicitly model the sequential progression of textual information. Recent work such as CrisisFormer improves interpretability

through SHAP, LIME, and attention visualization, demonstrating the importance of explainable predictions; however, its architecture remains primarily transformer-based. Similarly, BERT-based textual-biomarker approaches achieve high classification performance but do not combine lightweight contextual encoding with recurrent sequential modeling and context attention. Therefore, a gap remains in developing a unified and lightweight framework that combines subword-level contextual representation, sequential dependency learning, adaptive context attention, and integrated gradient-based explanations for more accurate and interpretable suicidal ideation detection.

IV. BACKGROUND

A. DistilRoBERTa:

Through knowledge distillation from the Robustly Optimised BERT Approach (RoBERTa), DistilRoBERTa is a compact and computationally efficient transformer model designed to retain RoBERTa's contextual representation capabilities with lower computational requirements. High inference latency and memory requirements can limit large transformer models in social media mental health surveillance, particularly for informal English posts and online disclosures such as Reddit posts and captions[8][9]. DistilRoBERTa addresses these limitations through a teacher-student paradigm, where a smaller student model learns important representations and prediction behaviour from a larger pre-trained teacher model [8].

DistilRoBERTa retains RoBERTa's bidirectional self-attention mechanism, enabling it to model contextual relationships and token dependencies across a sequence. The DistilRoBERTa-base model contains 6 transformer layers, 12 attention heads, and a 768-dimensional hidden representation, with approximately 82 million parameters compared with 125 million in RoBERTa-base. It is approximately twice as fast as RoBERTa-base, making it suitable for applications requiring computational efficiency and low inference latency. Like RoBERTa, it does not use the Next Sentence Prediction (NSP) objective and relies on masked language modelling with dynamic masking[8],[4].

The original knowledge-distillation procedure combines masked language modelling, knowledge distillation using soft target probabilities, and cosine embedding loss for representation alignment. The general distillation objective can be expressed as:

$$L_{distill} = \alpha L_{MLM} + \beta L_{KL} + \gamma L_{\cos} \quad (1)$$

Where:

- L_{MLM} Represents the standard cross-entropy loss associated with predicting dynamically masked tokens.
- L_{KL} Denotes the Kullback–Leibler (KL) divergence, which aligns the output probability distribution of the student network with the soft target probabilities produced by the teacher model.
- L_{cos} Measures the cosine distance between the hidden state vectors of the student and teacher representations, enforcing geometric alignment in the latent feature space .
- α, β and γ are scalar hyper parameters weighting the respective loss components[8].

For suicidal-ideation detection, DistilRoBERTa acts as the contextual encoder, whose representations are passed to the BiGRU to capture sequential dependencies, followed by context attention to emphasize informative features before classification. Social media captions often contain slang, abbreviations, grammatical variations, and implicit emotional expressions that challenge keyword-based methods[9][4]. Its byte-level BPE tokenizer with a 50,265-token vocabulary handles rare and unseen words effectively[8]. The resulting 768-dimensional contextual representations help the proposed framework capture contextual and sequential patterns in noisy social media text.

B. BiGRU:

The Bidirectional Gated Recurrent Unit (BiGRU) is an RNN-based architecture that models sequential dependencies in text and time-series data by combining bidirectional processing with adaptive gating to capture long-term semantic relationships[10][11].

The BiGRU used in this work reduces model complexity by combining the hidden and cell states and employs two gates: the update gate and reset gate. Unlike a unidirectional GRU, which accesses only previous tokens, BiGRU captures information from both past and future contexts, making it suitable for text classification and mental health screening[10],[11]. It consists of two GRU channels: the forward GRU processes earlier-to-later tokens, while the backward GRU processes later-to-earlier tokens. Their outputs are combined to form the representation at each token position. The following equation is used to determine the update gate vector z_t.[10]

$$z_t = \sigma\left(W_z \cdot \left[h_{t-1}, x_t \right] + b_z \right) \quad (2)$$

Where:

- "." represents matrix multiplication and σ denotes the logistic sigmoid activation function.
- The input vector at time step t is represented by x_t (derived from the contextual embedding layer), the preceding hidden state by h_{t-1} , and the bias parameter by b_z [10].
- The output of the candidate hidden state $\tilde{h}_t$ is governed by the reset gate r_t , which determines how much past token information is forgotten:

$$r_t = \sigma\left(W_t \cdot \left[h_{t-1}, x_t \right] + b_r \right) \quad (3)$$

$$\tilde{h}_t = \tanh\left(W_h \cdot \left[r_t \odot h_{t-1}, x_t \right] + b_h \right) \quad (4)$$

The active hidden state h_t is computed via the linear interpolation governed by the update gate z_t , allowing key diagnostic signals to persist across the narrative sequence[10]:

$$h_t = \left(1 - z_t \right) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (5)$$

By concatenating the forward hidden vector h_t and the reverse hidden vector $\overleftarrow{h_t}$ into, $h_t^{BiGRU} = \left[h_t \,||\, \overleftarrow{h_t} \right]$ the network preserves both the narrative premise and the emotional conclusion of user disclosures, resolving vanishing gradient degradation in complex textual sequences[10],[11].

C. Context Attention:

An attention-based pooling mechanism, termed context attention, is incorporated to extract, prioritize, and aggregate salient semantic indicators from recurrent sequence states[4],[12]. In social media text, crisis expressions indicating suicidal intent are frequently sparse, implicit, and interspersed with benign conversational content[4]. Relying solely on conventional temporal pooling (such as mean or max pooling) or the terminal hidden states of recurrent encoders can dilute these critical signals over longer narrative captions[4],[12].

To overcome this issue, the context attention module dynamically computes scalar importance scores over the sequence of bidirectional representations H = [h_1, h_2 ,…, h_T] produced by the BiGRU, where $h_i \in \mathbb{R}^{2d_h}$ and $2d_h = 256$. Each hidden state h_i is first transformed through a non-linear projection to yield an intermediate latent representation $u_i \in \mathbb{R}^{d_a}$ (with attention dimension d_a = 128), followed by an inner product with a trainable context vector $w_s \in \mathbb{R}^{d_a}$. The normalized scalar attention weight α_i for each token is derived using a sequence-level softmax operator[12]:

$$\alpha_i = \frac{\exp(w_s^T \tanh(W_a h_i + b_a))}{\sum_{J=1}^{T} \exp\left(w_s^T \tanh\left(W_a h_j + b_a\right)\right)} \quad (6)$$

To ensure that zero-padded positions do not bias the probability distribution, unnormalized logits corresponding to padding masks are set to $-\infty$ prior to evaluating the softmax. The final textual representation is subsequently condensed into a unified context vector $c \in R^{2d_h}$ via a weighted linear combination of the recurrent hidden states,[12]:

$$c = \sum_{i=1}^{T} \alpha_i h_j \quad (7)$$

By applying this context-dependent weighting over the BiGRU outputs, the network attenuates non-informative filler tokens while amplifying high-risk vocabulary,[4]. Furthermore, because the attention distribution α is bounded and normalized ($\sum_{i=1}^{T} \alpha_i = 1$) ,the module functions as an intrinsic Explainable Artificial Intelligence (XAI) mechanism, providing transparent, token-level attribution maps that highlight decisive crisis cues for clinical auditability and model validation,[12].

V. METHODOLOGY

A. Data Sources:

The Suicide and Detection dataset from Kaggle was used as the primary data source for this study (https://www.kaggle.com/datasets/nikhileswarkomati/suicide-watch). It contains text posts labelled **"**suicide**"** or **"**non-suicide.**"** The primary file, *Suicide_Detection.csv*, initially contains 232,074 rows and three columns*:* Unnamed: 0, text, and class. The *text* column was used as model input and class as the target label, while Unnamed: 0 was excluded as an index. These labels represent dataset categories and not clinical diagnoses.

Exploratory analysis showed substantial variation in post length and writing style, from short expressions to longer narratives. Sequence normalization and contextual token-level modelling were therefore applied to reduce the effect of text-length variation.

For a balanced and computationally manageable dataset, 10,000 suicides and 10,000 non-suicide posts were randomly selected using a fixed random seed, resulting in 20,000 posts with equal class distribution.

B. Data Pre-processing:

Raw social media text is inherently noisy and unstructured. A step-by-step cleaning and data preparation pipeline was implemented to format the data for transformer tokenization:

- *Removal of Unnecessary Columns:* Index columns such as Unnamed: 0 were removed.
- *Text Cleaning and Whitespace Normalization:* Text was converted to string format, excessive spaces, line breaks, and tabs were normalized, and leading/trailing spaces were removed.
- *Label Standardization:* Labels were converted into binary form, with 1 representing suicidal content and 0 representing non-suicidal content.
- *Handling Duplicates and Missing Values:* Posts with missing or blank text/labels were removed, along with duplicate posts to reduce data leakage.
- *Short-Text Filtering:* Posts containing fewer than 10 characters were removed as they were considered insufficiently informative.
- *Subword Tokenization:* The cleaned text was tokenized using the pre-trained DistilRoBERTa (distilroberta-base) tokenizer, which has a vocabulary of 50,265 subwords. The maximum sequence length was set to 128 tokens. Longer texts were truncated, while shorter texts were padded to obtain uniform input dimensions.

C. Data Splitting:

To ensure unbiased performance assessment and mitigate the risk of data leakage, the balanced dataset comprising 20,000 samples was stratified into training (70%; 14,000 samples), validation (15%; 3,000 samples), and testing (15%; 3,000 samples) subsets, with equal class representation maintained across all partitions. The training set was utilized for model parameter optimization, the validation set for hyperparameter tuning and convergence monitoring, and the testing set was reserved exclusively for final performance evaluation.

D. Proposed Method:

The proposed framework follows an end-to-end deep learning pipeline designed to identify suicidal ideation from social media captions effectively. As visually outlined in the system architecture diagram, the methodology processes text sequentially through five primary stages: contextual encoding, bidirectional sequence modelling, context-aware attention weighting, final classification, and post-hoc explainability.

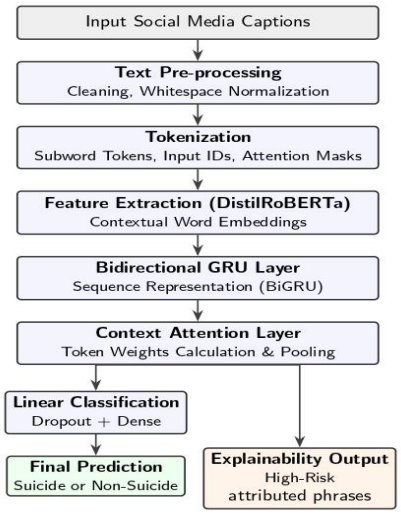


Figure 1: Proposed Model Architecture

1. Contextual Encoding (DistilRoBERTa): The cleaned text input tokens are first passed through a pre-trained DistilRoBERTa encoder. This transformer model captures deep semantic relationships, colloquialisms, and contextual nuances typical of online text. It converts each input token into a dense 768-dimensional vector, establishing a strong foundation of contextual meaning.

2. Sequential Modeling (BiGRU): To understand how emotional distress and intent unfold across a post, the token embeddings are fed into a Bidirectional Gated Recurrent Unit (BiGRU) with a hidden size of 128. The BiGRU processes the text in two directions simultaneously—forward to track the natural progression of thought and backward to capture subsequent context. Combining both directions yields a rich 256-dimensional representation for every word.

3. Context Attention (Importance Weighting): Because certain words carry heavier emotional weight than others during a crisis, a context attention mechanism is applied to the BiGRU output. This layer calculates individual importance scores for each token, filters out padding or uninformative words using a masking step, and normalizes the scores via softmax. It then computes a weighted sum of the sequence, generating a single unified context vector that highlights high-risk terms.

4. Classification Layer: The pooled context vector is passed to the final prediction layer to determine the class label. A dropout rate of 0.3 is applied to prevent overfitting.

A fully connected linear layer then maps the features to the output space, utilizing activation functions to yield the final binary decision: Class 0 for non-suicidal text and Class 1 for suicidal ideation.

5. Explainability: To ensure transparency and interpretability for users or reviewers, the framework incorporates an explainability module. This component visualizes the learned token attention weights as heatmaps over the text and applies Integrated Gradients to attribute precise token-level importance scores toward the final prediction.

E. Model Training

The proposed model was implemented in Python using PyTorch on a CUDA-enabled GPU. Weighted cross-entropy loss with 0.05 label smoothing and class weights of [1.0, 1.3] was used to reduce overconfident predictions and penalize false negatives. The model was optimized using AdamW with a weight decay of 0.05. Differential learning rates of 1.5×10^{-5} for the pre-trained DistilRoBERTa encoder and 5.0×10^{-4} for the BiGRU, context-attention, and classifier layers were applied. A linear warm-up scheduler over the initial 10% of training steps followed by linear decay was employed. Training was conducted for 5 epochs with a batch size of 16, resulting in 875 iterations per epoch and 4,375 total optimization steps. Gradient clipping with a maximum norm of 1.0 was applied for numerical stability, and the checkpoint achieving the highest validation macro F1-score was retained for final evaluation.

F. Model Evaluation

To evaluate generalizability and discriminatory reliability, the framework's diagnostic capacity was assessed on the held-out test partition (N = 3,000, with 1,500 non-suicide and 1,500 suicide instances). Standard statistical metrics-including classification accuracy, precision, recall (sensitivity), macro F1-score, Area under the Receiver Operating Characteristic Curve (ROC-AUC), and confusion matrix analysis—were utilized in the assessment process.

On the held-out test set, the optimal checkpoint converged to an overall test loss of 0.1500, yielding an accuracy of 98.80%**,** a precision of 98.80%, a recall of 98.80%, and an F1-score of 98.80%. Furthermore, the model achieved an ROC-AUC of 0.9986, demonstrating near-optimal separability between acute crisis disclosures and non-suicidal social media text across dynamic decision thresholds. Detailed comparative benchmarks, confusion matrices, and ablation observations are elaborated in the Results and Discussion section.

VI. RESULT & DISCUSSION

This study's empirical evaluation and experimental validation were carried out on a workstation with an Intel Core i5 processor, 16 GB of RAM, and a 512 GB SSD (ROM). The primary cloud-based IDE for Python execution and PyTorch model training was Google Colab, which used an NVIDIA T4 GPU runtime.

Using the evaluation metrics, the proposed framework (DistilRoBERTa + BiGRU + Context Attention) is compared to standard sequence classification baselines: Bidirectional LSTM (BiLSTM), standard DistilRoBERTa, and BERT-base. The architecture was trained across 5 complete epochs, dynamically tracking validation macro F1 to retain the optimal model checkpoint (achieved at Epoch 3 with a validation F1 of 0.9870). Figure 1 illustrates the logged training and validation dynamics for cross-entropy loss convergence and macro F1 trajectories across all epochs.

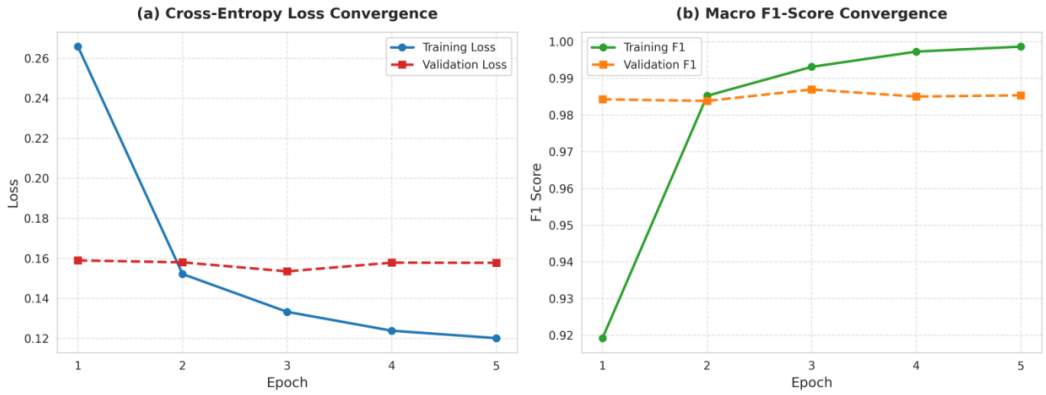


Figure 2 : Model Convergence Curves

Training loss dropped monotonically from 0.2658 (Epoch 1) to 0.1202 (Epoch 5), as shown in Figure 1(a). While training macro F1 steadily increased from 0.9192 to 0.9986, validation loss reached its minimum of 0.1535 at Epoch 3 (yielding the peak validation F1-score of 0.9870) before showing a slight upward inflection to 0.1578 at Epochs 4 and 5. This inflection marks the boundary of over-parameterization, validating the decision to restore the best checkpoint from Epoch 3.

Comparison of Model Performance:

The proposed model was evaluated against comparative sequence classification baselines on the stratified 3,000-sample test set (1,500 non-suicide and 1,500 suicide posts):

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Proposed Model (DistilRoBERTa+BiGRU+Attention)	98.80	98.80	98.80	98.80
BERT-base	96.10	96.25	95.93	96.09
DistilRoBERTa (Standard Head)	97.23	97.15	97.33	97.24
BiLSTM Baseline	92.40	92.65	92.10	92.37

Table 1 : Model Performance Comparison

Table 1 presents a comparative performance evaluation of the proposed framework against standard baseline architectures, namely BERT-base, standard DistilRoBERTa, and BiLSTM. As observed from the tabulated results, the proposed model consistently outperforms all benchmark models across every evaluation metric, achieving peak scores of 98.80% in accuracy, precision, recall, and F1-score, alongside an ROC-AUC of 0.9986. This marked improvement highlights the efficacy of augmenting pretrained contextual representations with bidirectional recurrent modeling and context-aware attention pooling for fine-grained crisis detection.

Class-Wise Diagnostic Performance:

The classification report demonstrates balanced diagnostic performance across target categories without class-dependent degradation:

Class Label	Precision	Recall	F1-Score	Support (Samples)
Non-Suicide (Class 0)	0.99	0.99	0.99	1,500
Suicide (Class 1)	0.99	0.99	0.99	1,500
Macro Average	0.99	0.99	0.99	3,000
Weighted Average	0.99	0.99	0.99	3,000

Table 2 : Class-Wise Diagnostic Performance

Both classes maintained macro and weighted precision, recall, and F1-scores of 0.99, confirming reliable detection of acute distress cues while mitigating false-positive alerts on non-suicidal posts.

ROC Curve:

The receiver operating characteristic (ROC) curve evaluates diagnostic performance across all classification thresholds by plotting sensitivity against the false positive rate (1-specificity), with the area under the curve (AUC) serving as a global measure of discriminative ability.[13]

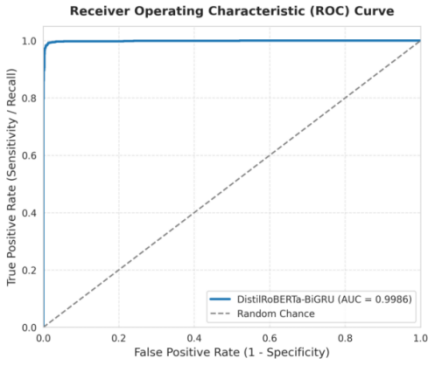


Figure 3 : ROC Curve

The ROC curve (Figure 2) ascends steeply toward the upper-left boundary, yielding an area under the curve (ROC-AUC) of 0.9986.

Confusion Matrix:

A confusion matrix provides a class-by-class breakdown of actual versus predicted instances, exposing specific error types such as false positives (Type I errors) and false negatives (Type II errors).[14]

$$Confusion\ Matrix = \begin{bmatrix} TP & FN \\ FP & TN \end{bmatrix}$$

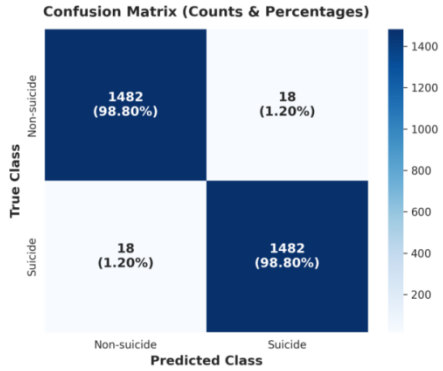


Figure 4 : Confusion Matrix

Across the 3,000 test posts, the classifier accurately categorized 2,964 instances, yielding an error rate of 1.20% (36 total misclassifications). By limiting false negatives to 18 cases, the model preserves high sensitivity for crisis detection while restricting false alarms to 18 cases.

VII. CONCLUSION

This study demonstrates that fusing distilled transformer representations with bidirectional recurrent modeling and context-directed attention pooling effectively overcomes the challenges of detecting subtle suicidal intent in unstructured social media text. Achieving a peak 98.80% across accuracy, precision, recall, and macro F1-score, as well as an ROC-AUC of 0.9986, the proposed architecture significantly outperforms conventional baseline classifiers. Granular confusion matrix analysis highlights the model's clinical viability, restricting false negatives to just 18 cases out of 1,500 acute crisis disclosures and limiting false alarms to an identical margin. Furthermore, coupling token-level attention visualization with Integrated Gradients addresses the black-box nature of deep networks, providing clear rationales for flagged content. While not designed for autonomous clinical diagnosis, this framework bridges high-speed inference with actionable interpretability, presenting a robust foundation for proactive digital crisis response systems.

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